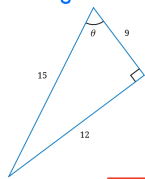


5.4

1) In triangle shown below we can find angle θ as follows



$$(a) \theta = \sin^{-1}\left(\frac{12}{15}\right)$$

$$(b) \theta = \cos^{-1}\left(\frac{9}{15}\right)$$

$$(c) \theta = \tan^{-1}\left(\frac{12}{9}\right)$$

2) Find exact value of each expression in radians

$$(a) \sin^{-1}\left(\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\pi}{4} \text{ rad}$$

$$(b) \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\pi}{4} \text{ rad}$$

$$(c) \tan^{-1}(1)$$

$$= \frac{\pi}{4} \text{ rad}$$

3) Use calculator to find approx value of expression

$$\sin^{-1}\left(\frac{7}{8}\right)$$

$$\text{radians: } \approx 1.065$$

$$\text{degrees: } \approx 61.045$$

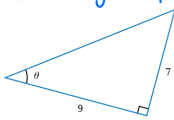
4) Use calculator to find approx value of expression

$$\cos^{-1}(6)$$

$$\text{radians: } \text{UNDEFINED}$$

$$\text{degrees: } \text{UNDEFINED}$$

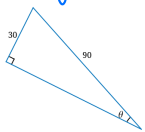
5) Find angle θ in degrees



$$\theta = \tan^{-1}\left(\frac{7}{9}\right)$$

$$\approx 37.9^\circ$$

6) Find angle θ in degrees



$$\theta = \sin^{-1}\left(\frac{30}{90}\right)$$

$$\theta \approx 19.5^\circ$$

7) Find all angles θ between 0° and 180° that satisfy given equation:

$$\cos(\theta) = \frac{5}{6}$$

$$\theta = \cos^{-1}\left(\frac{5}{6}\right)$$

$$\theta \approx 33.6^\circ$$

8) Find all angles θ between 0° and 180° that satisfy given equation:

$$\tan(\theta) = -16$$

Q II and ~~Q IV~~

$$\theta = \tan^{-1}(16)$$

$$\text{Q II: } \theta = 180^\circ - \theta$$

$$\approx 93.6^\circ$$

9) Find exact value of expression

$$\tan(\sin^{-1}(\frac{12}{13}))$$

$$\theta = \sin^{-1}(\frac{12}{13})$$

$$\sin \theta = \frac{12}{13}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$(\frac{12}{13})^2 + \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{5}{13}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{\frac{12}{13}}{\frac{\sqrt{5}}{\sqrt{13}}}$$

$$= \boxed{\frac{12}{5}}$$

10) Rewrite the expression as an algebraic expression in x

$$\sin(\tan^{-1}(x))$$

$$\theta = \tan^{-1}(x)$$

$$\tan \theta = x = \frac{x}{1} = \frac{\text{opposite}}{\text{adjacent}}$$

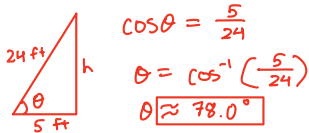
$$1^2 + x^2 = (\text{hypotenuse})^2$$

$$\text{hypotenuse} = \sqrt{1+x^2}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$= \frac{x}{\sqrt{1+x^2}}$$

11) 24-ft ladder is leaning against building. base of ladder is 5 ft from base of building. what is angle of elevation of ladder?



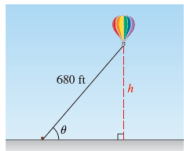
How high does ladder reach on building?

$$5^2 + h^2 = 24^2$$

$$h = \sqrt{24^2 - 5^2}$$

$$h \approx \boxed{23 \text{ ft}}$$

12) 680-ft rope anchors hot-air balloon



(a) Express angle θ as function of height h of the balloon

$$\sin \theta = \frac{h}{680}$$

$$\theta = \sin^{-1}(\frac{h}{680})$$

(b) Find angle θ if balloon is 630 ft high

$$\theta = \sin^{-1}(\frac{630}{680})$$

$$\theta \approx \boxed{67.9^\circ}$$

13) Find exact value of each expression in radians.

(a) $\sin^{-1}(-\frac{1}{2})$

$$= \boxed{-\frac{\pi}{6}} \quad \text{Q IV}$$

(b) $\cos^{-1}(-\frac{\sqrt{3}}{2})$

$$= \boxed{\frac{5\pi}{6}} \quad \text{Q II}$$

(c) $\tan^{-1}(-\frac{\sqrt{3}}{3}) \rightarrow \frac{-\sqrt{3}}{3} = -\frac{1}{\sqrt{3}}$



$$\boxed{-\frac{\pi}{6}}$$

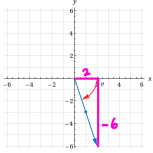
14) Use calculator to find approx value

$$\tan^{-1}(9)$$

$$\text{radians: } \boxed{\approx 1.460}$$

$$\text{degrees: } \boxed{\approx 83.660^\circ}$$

15) angle θ given in coordinate plane



find angle θ in degrees

$$\tan \theta = \frac{-6}{2}$$

$$\theta = \tan^{-1}\left(\frac{-6}{2}\right)$$

$$\boxed{\theta \approx -71.6^\circ}$$

16) angle θ given in coordinate plane



find angle θ in degrees

$$\tan \alpha = \frac{2}{-1}$$

$$\alpha = \tan^{-1}\left(\frac{2}{-1}\right)$$

$$\theta = 180^\circ - \alpha$$

$$\boxed{\theta \approx 153.4^\circ}$$

17) find all angles between 0° and 180° that satisfy given equation.

$$\sin \theta = \frac{1}{8}$$

Q I and Q II

$$\text{Q I: } \bar{\theta} = \sin^{-1}\left(\frac{1}{8}\right)$$

$$\boxed{\bar{\theta} \approx 7.2^\circ}$$

$$\text{Q II: } \theta = 180^\circ - \bar{\theta}$$

$$\boxed{\theta \approx 172.8^\circ}$$

18) find all angles between 0° and 180° that satisfy given equation.

$$\cos \theta = -\frac{4}{5}$$

Q II and ~~Q III~~

$$\bar{\theta} = \cos^{-1}\left(\frac{4}{5}\right)$$

$$\text{Q II: } \theta = 180^\circ - \bar{\theta}$$

$$\boxed{\theta \approx 143.1^\circ}$$

19) Find exact value of expression

$$\csc\left(\cos^{-1}\left(-\frac{8}{17}\right)\right)$$

$$\cos \theta = -\frac{8}{17}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta + \left(-\frac{8}{17}\right)^2 = 1$$

$$\sin \theta = \frac{15}{17}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$= \frac{1}{\frac{15}{17}}$$

$$= \boxed{\frac{17}{15}}$$

20) Find exact value of expression

$$\sec\left(\sin^{-1}\left(\frac{8}{17}\right)\right)$$

$$\sin \theta = \frac{8}{17}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{8}{17}\right)^2 + \cos^2 \theta = 1$$

$$\cos \theta = \frac{15}{17}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\sec \theta = \frac{1}{\frac{15}{17}}$$

$$\sec \theta = \boxed{\frac{17}{15}}$$

21) Find exact value of expression

$$\cos\left(\tan^{-1}\left(-\frac{15}{8}\right)\right)$$

$$\tan \theta = -\frac{15}{8}$$

$$8^2 + 15^2 = (\text{hypotenuse})^2$$

$$\text{hypotenuse} = 17$$

$$\cos \theta = \boxed{-\frac{8}{17}}$$

22) Rewrite the expression as an algebraic expression in x

$$\cot(\cos^{-1}(x))$$

$$\theta = \cos^{-1}(x)$$

$$\cos \theta = x = \frac{x}{1} = \frac{\text{adjacent}}{\text{hypotenuse}}$$

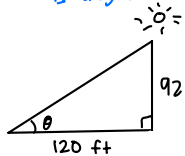
$$x^2 + (\text{opposite})^2 = 1^2$$

$$\text{opposite} = \sqrt{1-x^2}$$

$$\cot \theta = \frac{\text{adjacent}}{\text{opposite}}$$

$$= \frac{x}{\sqrt{1-x^2}}$$

23) 92-ft tree casts shadow that is 120 ft long. What is angle of elevation of sun?



$$\tan \theta = \frac{92}{120}$$

$$\theta = \tan^{-1}\left(\frac{92}{120}\right)$$

$$\theta \approx \boxed{37.5^\circ}$$